

SIMATS ENGINEERING

ECA09 - DIGITAL SIGNAL PROCESSING LAB

Experiment 08.1

Implementation and Analysis of Decimation and Interpolation in Multirate Signal Processing Using MATLAB.

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Aim

To implement and analyze the experiment described in the official ECA09 MATLAB lab manual.

Procedure / Calculations

Follow the experiment workflow given in the official lab manual: initialize MATLAB, define the specified signal/filter parameters, execute the algorithm, plot the required responses, and record the result.

MATLAB Program

```
clc; clear; close all;
% Original Sequence
n = 0:40;
x = sin(0.3*pi*n);
% Decimation by 2
y = x(1:2:end);
nd = 0:length(y)-1;
% Interpolation by 2
v = zeros(1,2*length(x));
v(1:2:end) = x;
ni = 0:length(v)-1;
figure; stem(n,x,'filled'); grid on; title('Original Sequence x[n]'); xlabel('n'); ylabel('Amplitude');
figure; stem(nd,y,'filled'); grid on; title('Decimated Sequence by 2'); xlabel('n'); ylabel('Amplitude');
figure; stem(ni,v,'filled'); grid on; title('Interpolated Sequence by 2'); xlabel('n'); ylabel('Amplitude');
figure; stem(n,x); hold on; stem(0:2:length(v)-2,v(1:2:end));
legend('Original x[n]', 'Interpolated Samples'); grid on;
title('Comparison of Sequences'); xlabel('n'); ylabel('Amplitude');
```

Output / Analysis

The output/analysis pages below are reproduced from the uploaded official ECA09 MATLAB lab manual for this experiment.

```
nd = 0:length(y)-1;
```

```
% Interpolation by 2
```

```
v = zeros(1,2*length(x));
```

```
v(1:2:end) = x;
```

```
ni = 0:length(v)-1;
```

```
% Figure 1: Original Sequence
```

```
figure;
```

```
stem(n,x,'filled');
```

```
grid on;
```

```
title('Original Sequence x[n]');
```

```
xlabel('n');
```

```
ylabel('Amplitude');
```

```
% Figure 2: Decimated Sequence
```

```
figure;
```

```
stem(nd,y,'filled');
```

```
grid on;
```

```
title('Decimated Sequence by 2');
```

```
xlabel('n');
```

```
ylabel('Amplitude');
```

```
% Figure 3: Interpolated Sequence
```

```

figure;
stem(ni,v,'filled');
grid on;
title('Interpolated Sequence by 2');
xlabel('n');
ylabel('Amplitude');

```

% Figure 4: Comparison

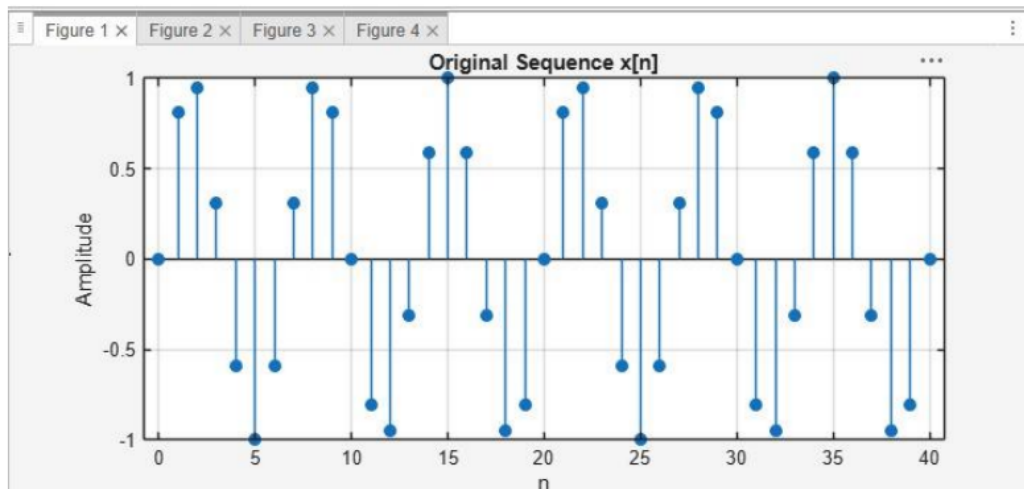
```

figure;
stem(n,x,'b'); hold on;
stem(0:2:length(v)-2,v(1:2:end),'r');
legend('Original x[n]','Interpolated Samples');
grid on;
title('Comparison of Sequences');
xlabel('n');
ylabel('Amplitude');

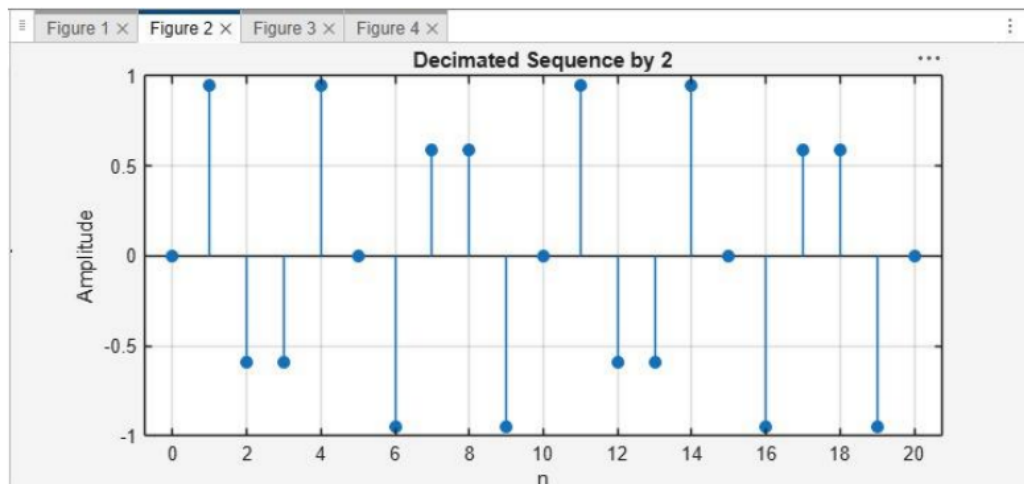
```

Output:

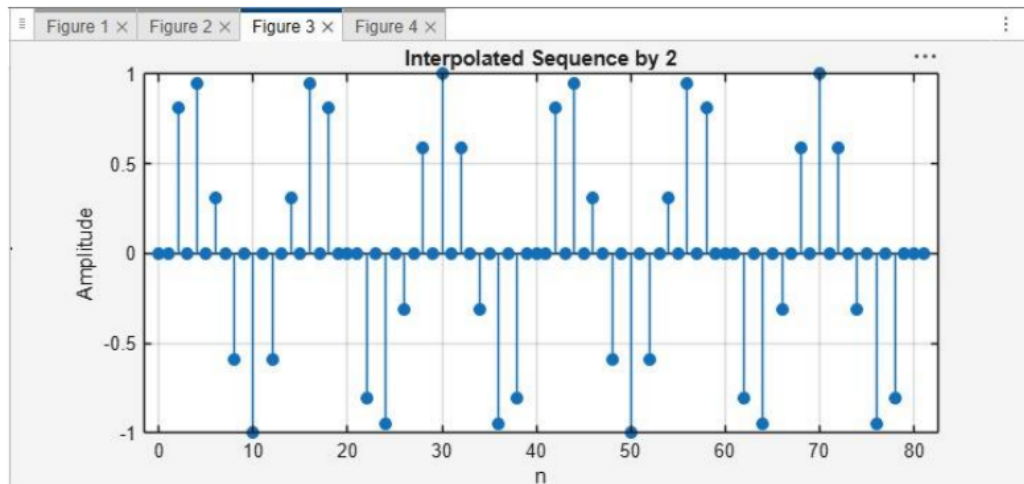
Objective 1:



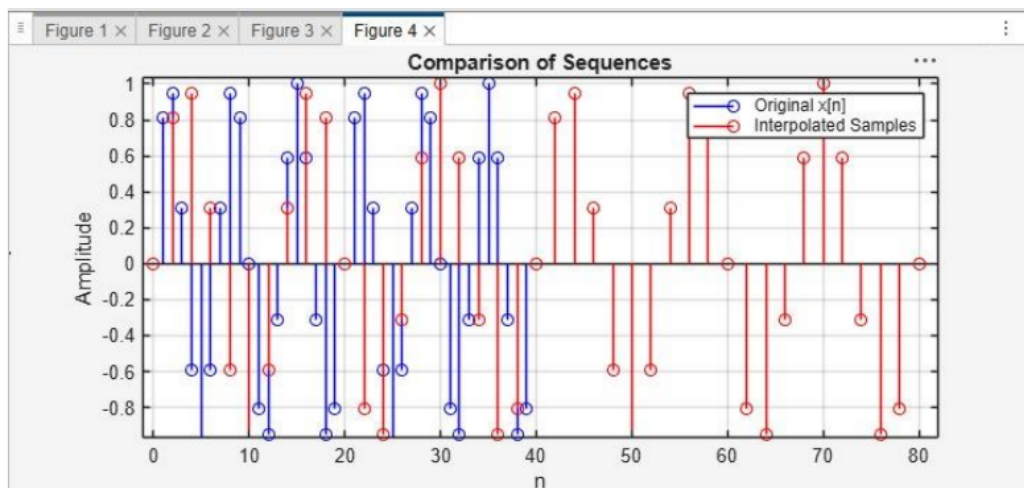
Objective 2:



Objective 3:



Objective 4:



Result :

1. A discrete-time sinusoidal sequence was successfully generated for multirate processing.
2. The discrete-time sequence was successfully decimated by a factor of 2, reducing the number of samples and the sampling rate.
3. The discrete-time sequence was successfully interpolated by a factor of 2, increasing the number of samples and the sampling rate.

4.. The effects of downsampling and upsampling were successfully analyzed by comparing the original, decimated, and interpolated signals.

Practical Question 1

Generate a discrete-time sinusoidal signal of frequency **500 Hz** sampled at **8 kHz**. Perform decimation by a factor of 3 and interpolation by a factor of 3 using MATLAB.

Tasks:

1. Generate the discrete-time sinusoidal sequence.
2. Perform decimation by a factor of 2.
3. Perform interpolation by a factor of 2 on the decimated sequence.
4. Plot and compare the original, decimated, and interpolated signals.

Aim (10)	Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

Practical Question 2

Generate a discrete-time sinusoidal signal with a frequency of **1 kHz** and a sampling frequency of **10 kHz**. Implement multirate processing by performing decimation and interpolation with a factor of 2.

Tasks:

1. Generate the input sinusoidal sequence.
2. Decimate the sequence by a factor of 2.
3. Interpolate the decimated sequence by a factor of 2.
4. Compare the original, decimated, and reconstructed signals and comment on the effect of multirate processing.

Aim (10)	Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

Result

To demonstrate sample-rate reduction by decimation and sample-rate increase by interpolation, both by a factor of 2.

Student Details

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